

RAKSHIT KUMAR

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Education

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science | CGPA: 9.20

Aug 2023 – May 2027

Chennai, India

Experience

Indian Institute of Information Technology Allahabad

May 2025 – July 2025

Research and Development Intern [\[GitHub\]](#)

On-site

- Developed an **edge-AI system** on **Raspberry Pi 5** with **AI HAT+ (13 TOPS)** for task driven object detection using **TFLite SSD-MobileNet** and **YOLOv11** models.
- Integrated **LLMs** (Phi-2, Gemma, Qwen) via **llama.cpp** with **Chain-of-Thought (CoT)** prompting to reason about object suitability, **increasing model accuracy from 52% to 84%**.
- Built a real-time **PiCamera2 (OpenCV) + Vosk speech system** for multimodal interaction and filtered bounding boxes using LLM responses.
- Benchmarked LLMs across **6 metrics**, monitored system performance with **nmon** and **Matplotlib**, and plotted comparative graphs, observed that **Phi-2 reduced inference latency by ~20 seconds compared to Gemma and Qwen** on Raspberry Pi 5.

Elevance Skills

Dec 2025 – June 2026

Full Stack Web Development Intern [\[GitHub\]](#)

Remote

- Developed a full-stack social networking web application focused on real-time user interaction, with **secure authentication, role-based access control**, and **scalable content interaction workflows** using **Next.js, React, TypeScript, Node.js, Express.js, and MongoDB**.
- Built enterprise-style security and user management features including **OTP-based login, Google OAuth, Firebase authentication, device tracking, login history monitoring**, and **browser-based access control**.
- Designed scalable backend APIs and integrated advanced functionalities such as **Multilingual UI support, Multimedia uploads, Browser notifications, Subscription plans**, and **payment/media services** using **Razorpay and Cloudinary**.

Projects

AI Interviewer [\[GitHub\]](#) [\[Live\]](#)

June 2026

- Built a full-stack AI Interviewer platform using **MERN, FastAPI, Socket.IO, Gemini, Whisper, and Chart.js**, with secure authentication and automated coding and voice-based interviews across 3 interview levels and 20 technical domains.
- Integrated **deep-learning-based speech-to-text** using Whisper and **LLM-powered evaluation** using Gemini to automate oral and coding assessment, delivering **structured feedback with technical score, confidence score, and answer analysis** for every response. reducing human effort by 100%.
- Optimized the interview pipeline by redesigning AI architecture from **O(N+1)** Gemini calls to **O(2)** calls per interview where N is the number of questions, drastically reducing API usage and improving system efficiency.

Integrated Flood Prediction & Smart Evacuation System

Nov 2025 – March 2026

- Developed an end-to-end flood prediction and GIS-based evacuation routing system using **XGBoost, LSTM, Python, and QGIS**, trained on **10+ years of rainfall, runoff, and soil moisture data**, with **Chennai as the primary study region**.
- Generated flood risk zonation using the **HAND model** and dynamically filtered safe shelters for disaster response planning.
- Implemented network-based shortest path optimization to compute optimal evacuation routes during flood scenarios.

Stock Market Price Predictor [\[Github\]](#) [\[Live\]](#)

July 2025 – Sep 2025

- Developed and deployed an AI-powered stock market prediction platform using **Python, Streamlit, Scikit-learn, Plotly, Yahoo Finance API, VADER, and TextBlob**, benchmarking **5 machine learning models** using **RMSE, MAE, and R²**.
- Engineered **20+ technical and sentiment-driven features** from **2 years** of historical market data and **35+** financial news headlines to generate next-day stock price forecasts and investment insights.
- Built an end-to-end analytics dashboard with real-time market visualization, model comparison, sentiment analysis, and investment simulation.

Skills

Languages: HTML, CSS, Tailwind CSS, JavaScript, C, C++, Python, TypeScript, Python

Frameworks/Libraries: React.js, Node.js, Express.js, Redux, OpenCV, NumPy, Next.js, Streamlit, Scikit-learn, Pandas, Plotly

Technical Skills: Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Machine Learning

Tools/Databases: Git, GitHub, MySQL, MongoDB, Firebase, Cloudinary

Soft Skills: Analytical Thinking, Problem Solving, Communication, Leadership, Collaboration, Adaptability, Business Acumen, Entrepreneurship

Achievements and Leadership

- Award of Excellence during IIT Allahabad internship**, honored for outstanding contribution in building an edge-AI task driven object detection system with Raspberry Pi 5, AI HAT+ (13 TOPS) and LLM integration.
- Founder, DRIFTZ Clothing Brand (Business)** - Built and managed DRIFTZ, a premium streetwear clothing brand, achieving **sales of 3 lakhs+ rupees** and successfully **selling 400+ oversized T-shirts**. Oversaw product design, branding, marketing, customer targeting, and business operations from concept to execution.
- Technical Head, Liftoff Club** - Led web design and development initiatives for the club, enhancing digital engagement while mentoring developers through code reviews and technical guidance. Collaborated with cross-functional clubs to organize workshops, technical events, and learning sessions, fostering skill development across members.